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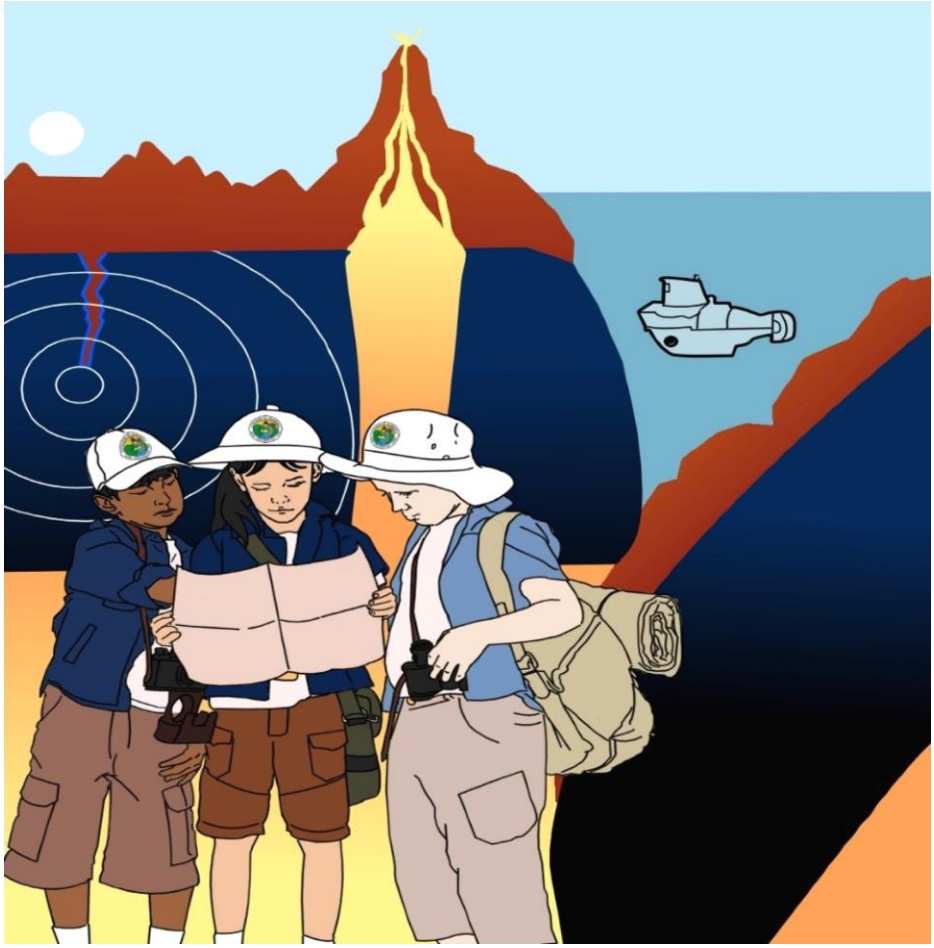
Department of Education
National Capital Region
SCHOOLS DIVISION OFFICE
MARIKINA CITY

Science

Quarter 1 – Module 3

Processes and Landforms along Plate Boundaries



Guiller P. Belen

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What I Need to Know

The purpose of this module is to help you understand the processes and landforms along plate boundaries. The module is divided into five lessons, namely:

- Lesson 1 – Convergence of Oceanic and Continental Plates
- Lesson 2 – Convergence of Two Oceanic Plates
- Lesson 3 – Convergence of Two Continental Plates
- Lesson 4 – Divergence of Two Plates
- Lesson 5 – Processes along Transform Boundaries

After going through this module, you are expected to **explain the different processes that occur along the plate boundaries. S10ES – Ia-j-36.3)**

Specifically, you are expected to:

1. explain what happens when oceanic plate and continental plate collide;
2. identify the landforms associated with oceanic and continental plate convergence;
3. explain what happens when two oceanic plates collide;
4. identify the landforms associated with oceanic plate convergence;
5. explain what happens when two continental plate boundaries collide;
6. identify the landforms associated with continental plate convergence;
7. explain the processes that occur along divergent boundaries;
8. identify the landforms associated with divergent plate boundaries;
9. explain the processes that occur along transform boundaries, and
10. identify the landforms associated with transform boundaries.

What I Know

Read and understand each item carefully and encircle the letter of the correct answer.

1. What happens when oceanic and continental plate move towards each other?
   A. The oceanic plate moves below the continental plate.
   B. The oceanic plate moves above the continental plate.
   C. The oceanic plate and continental plate fold upwards.
   D. The oceanic plate and continental plate fold sideways.

2. Which term refers to the moving of one plate beneath the other plate due to the difference in density?
   A. faulting
   B. folding
   C. subduction
   D. uplift

3. What geologic feature is formed on the surface directly above the subduction zone due to the rising of molten material?
   A. mountains
   B. trench
   C. valley
   D. volcano

4. What happens when two oceanic plate moves towards each other?
   A. The denser oceanic plate moves below the less dense oceanic plate.
   B. The denser oceanic plate moves above the less dense oceanic plate.
   C. Both plates fold upward.
   D. Both plates fold sideward.

5. Which is produced by the convergence of continental and oceanic plates?
   A. Ocean ridge
   B. Rift valley
   C. Trench
   D. High mountain ranges

6. What geologic feature is formed when a continental plate converges with another continental plate?
   A. Volcano
   B. Mountain ranges
   C. Deep-sea trenches
   D. Volcanic Island Arc

7. What plates formed the Philippine Island?
   A. Eurasian plate and Indian Plate
   B. Eurasian plate and Pacific Plate
   C. Philippine Sea plate and Pacific Plate
   D. Philippine Sea Plate and Eurasian Plate

8. Which two plates formed the Himalayas Mountain Ranges?
   A. African Plate and Eurasian Plate
   B. Pacific Plate and Eurasian Plate
   C. Eurasian Plate and Indian Plate
   D. South America Plate and Nazca Plate

9. Why do two continental plates uplift when it converges?
   A. Both plates are heavy and dense.
   B. Either of the plates is heavy and dense.
   C. Both plates are light and less dense.
   D. Either of the plates is light and less dense.

10. Which event is most likely to take place first during the convergence of oceanic plate and continental plate?
   A. Earthquake
   B. Tsunami
   C. Mountain building
   D. Volcanic eruption

11. Which is characterized by plates that move away from each other?
   A. Divergent plate boundary
   B. Transform plate boundary
   C. Oceanic – oceanic Convergent plate boundary
   D. Continental – continental Convergent plate boundary

12. Which term refers to the landform produced when two continental plates move away from each other?
   A. Rift valley
   B. Ocean ridges
   C. Trench
   D. Sinkhole

13. How will you describe the rocks in the mid-ocean ridge?
   A. It forms a shallow valley
   B. It is youngest in the middle of the ridge.
   C. The center is relatively cooler compared to the surrounding area.
   D. The rocks at the ridges change its composition when it moves away.

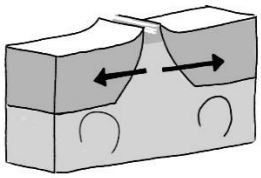
14. What plates formed the Mid-Atlantic Ridge?
   A. South American Plate and Nazca Plate
   B. African Plate and Eurasian Plate
   C. African Plate and South American Plate
   D. Eurasian plate and South American Plate

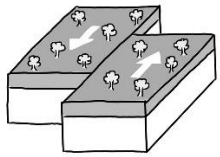
15. What is formed in a transform boundary?
   A. Mountain
   B. Volcano
   C. Faults
   D. Island

Lesson 1	Convergence of Continental and Oceanic Plate

What's In

Name and describe the following types of plate boundary movements.





Boundary Movement:

1._____

_____

Boundary Movement:

2._____

_____

Boundary Movement:

3._____

_____

What's New

Activity: Convergence of Oceanic and Continental Plates

Objectives:

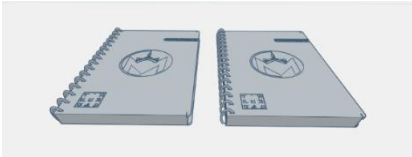
1. Explain what happens when oceanic plate and continental plate collide
2. Identify the landforms associated with oceanic and continental plate convergence.

You Will Need:

two notebooks (equal size)

Procedure:

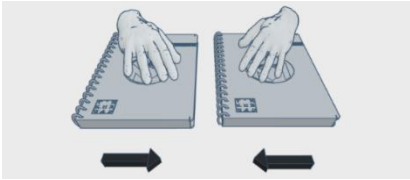
1. Lay the two notebooks flat (side to side and same orientation) on a flat surface.



Illustrated by Keith Angeline N. Alejandro

2. Hold the top of the notebooks and gently move them towards each other.

3. Observe what happens. Write your observations and answers to the guide questions.



Illustrated by Keith Angeline N. Alejandro

Guide Questions:

1. Which notebook (left or right) represents the oceanic plate? How did you say so?

2. Which notebook (left or right) represents the continental plate? How did you say so?

3. What happens to the left notebook as it moves toward the right notebook? What might be the reason for this?

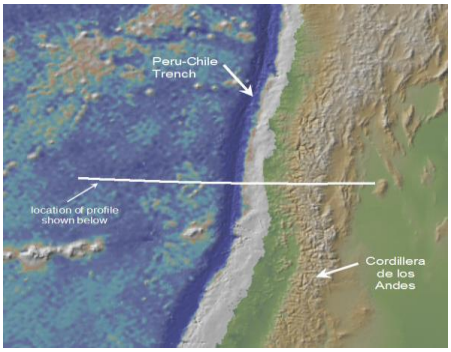
4. What will happen to the plates if the collision continues to happen?

5. What events might occur during this process?

6. What landforms are formed?

What Is It

Activity 1 showed the **collision** of the *oceanic plate* and *continental plate*. You noticed that as they move towards each other, one moved below the other. Since the notebook's backbone is more compact, it tends to move below the other notebook which has a loose end. This can be related to the **density** of the two plates. The oceanic plate is *denser* than the continental plate thus oceanic plate moves below the continental plate. This process is called **subduction,** where the denser plate tends to move below the other plate. As the denser plate moves down it creates a gap that runs deep in the sea called a **trench**. One example is the *Peru-Chile Trench* where the



Nazca-South American Margin Bathymetry and topography (created with GeoMapApp) by BRIAN ROMANS is licensed under CC BY-NC-ND

Figure 1: Formation of Peru-Chile Trench and Andes Mountain range due to convergent plate boundary movement

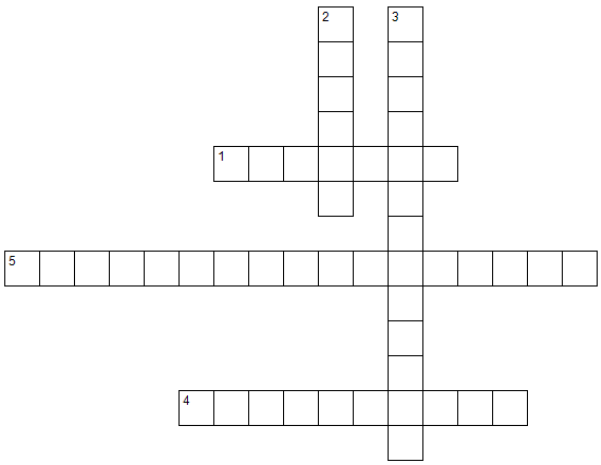
Nazca plate (oceanic plate) subducts underneath the *South American Plate* (continental plate).

Also, in the activity, you have seen that the loose side of the notebook folded upwards. This led to the formation of **mountains**. An example is *Andes Mountain Range* which runs parallel along *Peru-Chile Trench.*

Over millions of years, as the oceanic plates move down the mantle, it melts. The *high temperature* in the **mantle** causes the rocks to break and melt. This will eventually rise and cause volcanic activities on the continental plate above it.

What's More

Answer the crossword below. Write your answers on a separate sheet of paper.



Across:

1. vent or opening in the earth's surface
4. moving of plate under another
5. less dense

Down:

2. deepest part of the ocean
3. dense

What I Have Learned

Complete the statement. Write your answers on a separate sheet of paper.

A. Different geologic processes occur in a convergent plate boundary. These processes produce different *(1.)* _____and geologic events.

B. In a *(2.)*_____, the oceanic plate moves under the continental plate due to its density. The process is called *(3.)*_____, and this led to the formation of *(4.)*_____ between them. Over the continental plate, *(5.)* _____ were built. After the continued collision, the oceanic crust will reach the mantle where the temperature is high enough to melt the plate. Eventually, this will rise and form a volcano. In addition, shallow and deep earthquakes are also generated, which may lead to the production of (*6.)* _____ in the ocean.

What I Can Do

Draw a diagram of continental-oceanic plate boundaries and identify the different landforms formed due to this movement. Draw and label your answers.

Assessment

Write TRUE if the statement is correct. If the statement is wrong, change the underlined word/s to make the sentence correct.

_____1. The oceanic plate moves below the continental plate because of its density.
_____2. A trench is the shallow part of the gap between the oceanic plate and continental plate.
_____3. Mountains are built over the continental plate.
_____4. Volcanoes are formed over the oceanic plate.
_____5. Oceanic-continental plate convergence forms Mt. Everest.

Additional Activities

Performance Task. Research on any famous geologic landform that was created by **convergent oceanic – continental plate boundary movement** and create an infographic showing its location, origin, and other interesting facts. It may be in a form of physical or digital output. (Infographic is a visual image used to represent information.) Rubrics for scoring is found on page 13.

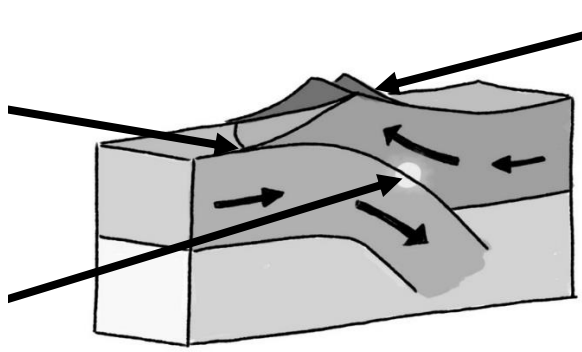
Lesson 2	Convergence of Two Oceanic Plates

What's In

Identify the processes/ landforms along the continental-oceanic plate boundary.

3._____

2._____



1._____

What's New

Activity: Convergence of Two Oceanic Plates

Objectives:

1. Explain what happens when two oceanic plates collide, and;
2. Identify the landforms associated with oceanic plate convergence.

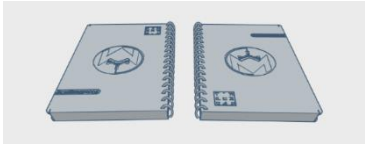
You Will Need:

two notebooks (equal size)

Procedure:

1. Lay the two notebooks flat (side to side and rotate the left notebook so that the backbones will face each other) on a flat surface.

2. Hold the top of the notebooks and gently move them towards each other.



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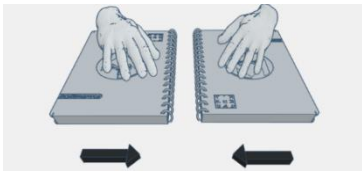

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3. Observe what happens. Write your observations and answers to the guide questions.

Guide Questions:

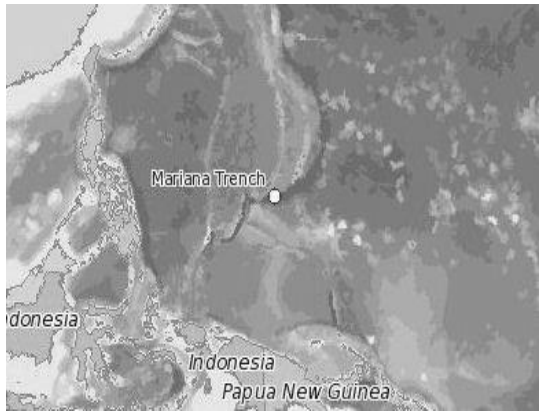
1. What happens to the left notebook as it moves towards the right notebook? What might be the reason for this?

2. What will happen to the plates if the collision continues to happen?

3. What events might occur during this process? What landforms are formed?

What Is It

Activity 2 depicted the **collision** of *two oceanic plates*. From the activity, you noticed that one notebook subducted under the other. This shows how the oceanic plate moves when it collides with other oceanic plates. **Subduction** takes place because there is still a difference in the **density** between the two plates. The older oceanic plate is denser than the young oceanic plate. The *denser* plates move below the plate. This also leads to the formation of a **deep ocean trench**.



Mariana trench location by Planiglobe is licensed under CC BY-NC-ND

Figure 2: Marianas Trench found in between the Pacific plate and Philippine Sea plate

Marianas Trench is formed due to the subduction of the *Pacific Plate* (oceanic plate) under the less dense *Philippine Sea Plate* (oceanic plate). In addition, these trenches become the source of an **underwater earthquake** that has enough energy to generate a **tsunami**.

When the subduction continues, undersea mountains will become undersea volcanoes. This undersea volcano will reach the surface of the sea, thus, creating a **volcanic island arc**. *Philippines* and *Japan* are examples of these volcanic island arc.

What's More

Rearrange the jumbled words related to the convergence of two oceanic plates.

_____1. SBUTUOCDIN – moving of one plate below another plate
_____2. NCTRHE – deep part of the ocean
_____3. AICNVOLC SINDLA CAR – an island formed by an undersea volcano
_____4. NASTUMI – displacement of water in the ocean
_____5. OSICLLION – moving of plates toward each other

What I Have Learned

Complete the statement.

In a *(1.)* _____, the denser plate subducts under the less dense plate. This leads to the formation of *(2.)* _____, undersea mountains which may eventually lead to *(3.)* _____ due to the melted subducted plate. Over millions of years, the undersea volcano may surface over the sea forming *(4.)* _____.

What I Can Do

Compose a poem or song on the processes involved in oceanic–oceanic plate boundary movement and include the different landforms formed.

Rubrics for scoring

Content (10 pts) Accurate and detailed information is provided.

Organization (5 pts) Information are systematically organized, and it is easy for the reader to understand.

Assessment

Write TRUE if the statement is correct. If the statement is wrong, change the underlined word/s to make the sentence correct.

_____1. When two oceanic plates meet, the younger oceanic plate subducts.
_____2. The older oceanic plate is denser than the younger oceanic plate.
_____3. Tsunamis are formed over the ocean due to the sudden release of energy from moving plates underwater.
_____4. Philippines and Japan are examples of a volcanic island arc.
_____5. Marianas trench is formed because the Philippine Sea Plate subducts under the Pacific plate.

Additional Activities

Performance Task. Research on any famous geologic landform that was created by **convergent oceanic – oceanic plate boundary movement** and create an infographic showing its, location, origin, and other interesting facts. It may be in a form of physical or digital output.

Rubrics for scoring

Content (10 pts) Accurate and detailed information is provided.

Organization (5 pts) Information are systematically organized, and it is easy for the reader to understand

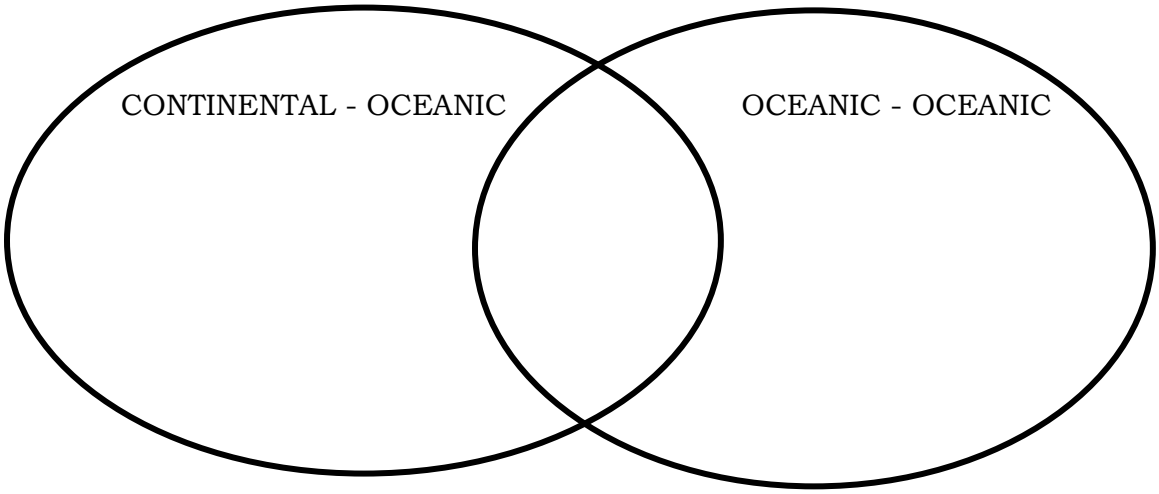
Visual Appeal (5 pts) Letter, colors, images, layouts, and other visual elements help in the expressing overall idea of the infographics

Citation (5 pts). All sources of information are cited using APA style.

Lesson 3	Convergence of Two Continental Plates

What's In

Compare and contrast the convergence between continental and oceanic plates and the convergence of two oceanic plates using a Venn diagram.



What's New

Activity: Convergence of Two Continental Plates

Objectives:

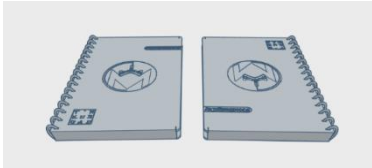
1. Explain what happens when two continental plates collide,
2. Identify the landforms associated with continental plate convergence.

You Will Need:

two notebooks (equal size)

Procedure:

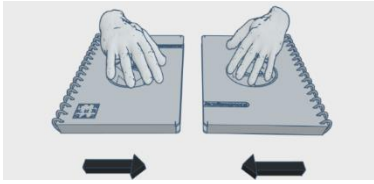
1. Lay the two notebooks flat (side to side and rotate the right notebook so that the opening side of the notebooks faces each other) on a flat surface.
2. Hold the top of the notebooks and gently move them



Illustrated by Keith Angeline N. Alejandro

towards each other.

3. Observe what happens. Write your observations and answers to the guide questions.



Illustrated by Keith Angeline N. Alejandro

Guide Questions:

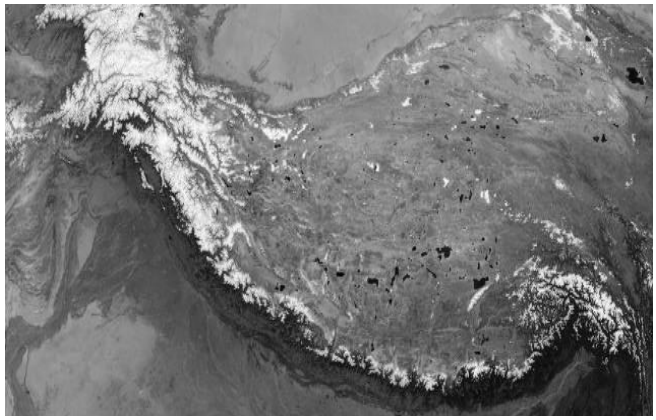
1. What happens to the left notebook as it moves towards the right notebook? What might be the reason for this?

2. What will happen to the plates if the collision continues to happen?

3. What events might occur during this process? What landforms are formed?

What Is It

Activity 3 exhibited the **collision** of **two continental plates**. Since the edge of the notebooks is loose, both sides move up. This indicates that when a *less dense* continental plate collides with another continental plate, the two *uplifts* and *folded* upwards. This leads to the formation of **mountain ranges**. An example is the convergence of the *Eurasian Plate* (continental plate) and *Indian Plate* (Continental plate) which formed the *Himalayan mountain ranges.*



Himalayas landsat 7 by NASA is licensed under CC BY-SA

Figure 3. The Himalayan Mountain ranges.

As the plates continue to move towards each other, the mountain continues to rise, and this explains the current height of *Mt. Everest*. This kind of collision only produces **shallow earthquakes.**

What's More

Look for six words related to the convergence of two continental plates. Use these to create a 2-3 sentence paragraph, explaining the processes and landforms involved in this movement.

_____
_____
_____
_____
_____
_____
_____

M	L	E	X	L	R	X	E	V	E	R	E	S	T	D
O	N	D	K	J	I	C	O	L	L	I	S	I	O	N
G	M	I	R	Q	N	D	K	N	Q	F	W	Y	H	P
D	B	T	A	X	G	M	A	L	I	K	R	I	T	B
W	K	L	G	T	I	Z	S	Z	U	Z	M	U	C	A
S	H	B	N	B	N	F	E	R	W	A	E	K	D	V
L	J	U	I	J	L	U	G	I	L	Q	A	G	A	M
O	Y	Z	D	C	H	M	O	A	U	N	R	P	F	L
J	C	G	L	V	X	E	Y	M	Y	V	T	E	K	S
B	T	M	O	Z	D	A	U	F	G	X	H	C	G	F
C	Y	C	F	G	S	R	Z	X	J	C	Q	E	H	I
N	R	O	Z	X	Z	V	U	W	O	E	U	X	D	M
Y	J	K	W	R	P	N	T	B	T	M	A	Q	H	O
Q	N	H	J	K	B	I	H	N	V	X	K	J	Z	K
D	T	G	Q	X	C	M	T	C	J	I	E	U	V	L

What I Have Learned

Complete the statement.

In a *(1.)*_____, the two plates are uplifted which creates *(2.)*_____. Since the two plates are *(3.)* _____, neither is subducted instead the plate folds and were pushed upwards forming high mountains. Also, this is accompanied by shallow *(4.)* _____.

What I Can Do

Create a comics strip on how Mt. Everest is formed. You can research the history of its formation.

Rubrics for scoring

Content (10 pts) Accurate and detailed information is provided.

Organization (5 pts) Information are systematically organized, and it is easy for the reader to understand

Visual Appeal (5 pts) Letter, colors, images, layouts, and other visual elements help in the expressing overall idea of the infographics

Citation (5 pts). All sources of information are cited using APA style.

Assessment

Write TRUE if the statement is correct. If the statement is wrong, change the underlined word/s to make the sentence correct.

_____1. Plates fold and uplift when two less dense continental plates converge.
_____2. Deep earthquake takes place in a continental-continental plate convergence.
_____3. The volcano is formed due to the collision of two continental plates.
_____4. Mt. Everest is the tallest mountain on the Earth.
_____5. Himalayan mountain ranges are formed due to the convergence of the Eurasian plate and Indian plate.

Additional Activities

Performance Task. Research on any famous geologic landform that was created by **convergent continental – continental plate boundary movement** and create an infographic showing its, location, origin, and other interesting facts. It may be in a form of physical or digital output.

Rubrics for scoring

Content (5 pts) Accurate and detailed information is provided.

Organization (5 pts) Information are systematically organized, and it is easy for the reader to understand

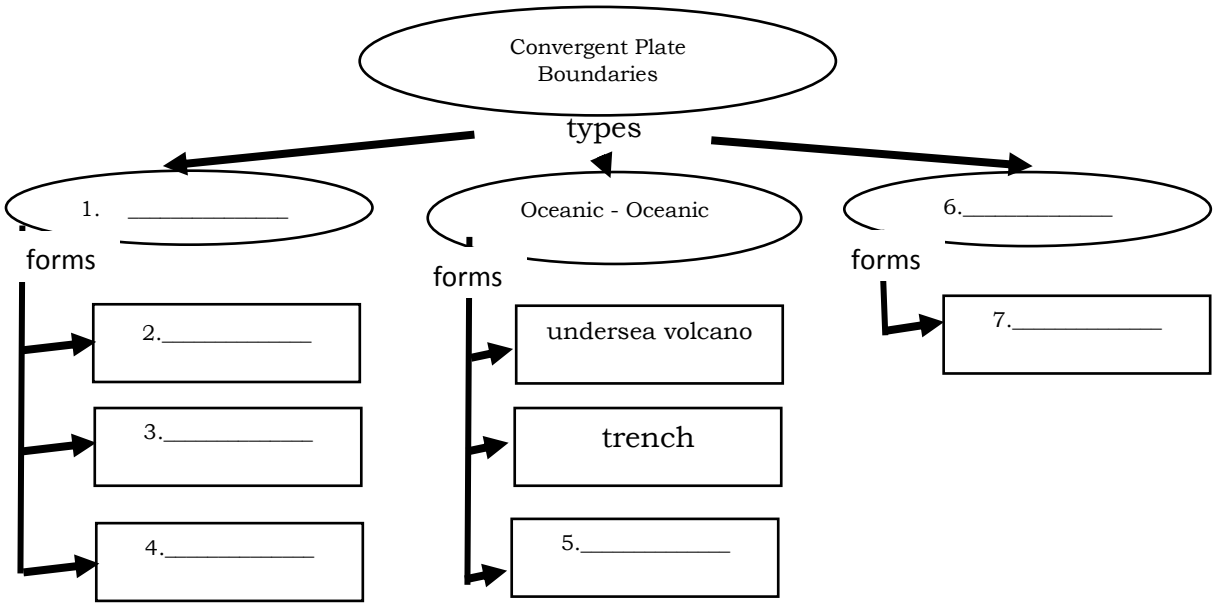
Visual Appeal (5 pts) Letter, colors, images, layouts, and other visual elements help in the expressing overall idea of the infographics

Citation (5 pts). All sources of information are cited using APA style.

Lesson 4 Divergence of Two Plate

What's In

Complete the diagram about convergent plate boundary.



What's New

Activity: They Move Apart

Objectives:

1. Explain the processes that occur along divergent boundaries, and;
2. Identify the landforms associated with divergent plate boundaries.

You Will Need:

folder
pair of scissors
1 short or long bond paper
Coloring material (preferably green, blue, and red)

Procedure:

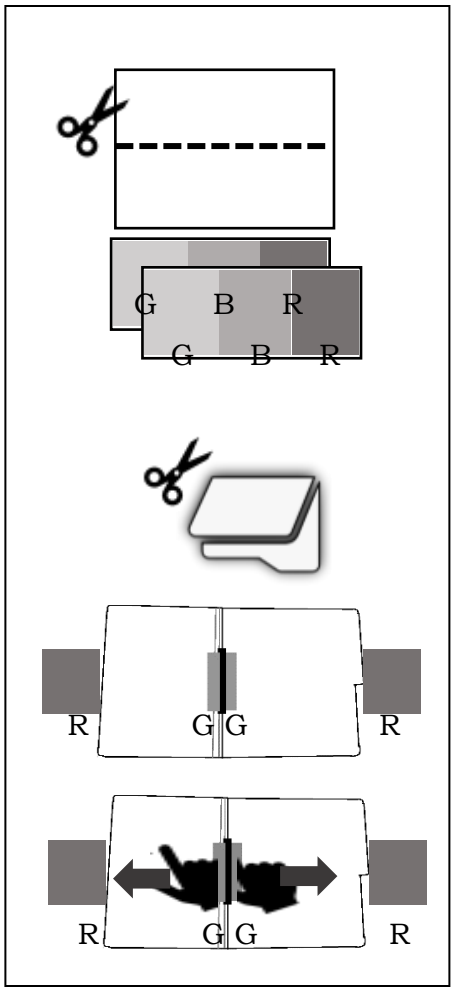
1. Cut the bond paper into two long strips.

2. Fold both papers into three to create three sections, then color it with green, blue, and red, respectively. (Start from-the left section to the right if it is oriented landscape.)

3. Make a slit on a folder. The length of the slit should be a little longer than the shorter end of the strip of paper you have made.)

4. Insert the two strips of paper starting with the green part. The colored part should be facing each other.

5. Slowly pull the paper away from each other until the end of the red part. Observe what happens. Write your observations and answers to the guide questions on a separate sheet of paper.



Illustrated by Guiller Belen

Guide Questions:

1. What does each colored part represent?

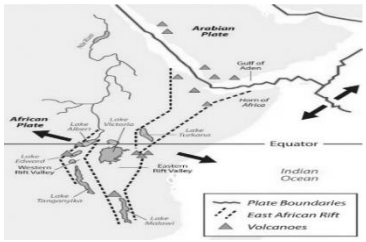
2. What have you noticed on the paper when you start pulling the strips away from each other?
   What does it indicate?

3. What is formed by this kind of movement?

What Is It

The **divergent plate boundary** is the boundary that allows two plates to move away from each other. It may be found in continents and oceans. It is the reason why we have a new crust and new oceans.

In the prior activity, you simulated movement in a divergent plate boundary. The colored part represented the plates, the bodies of water, and the magma inside the earth.

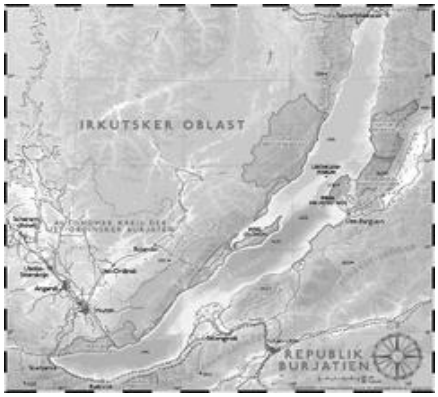


Great Rift Valley of East Africa by USGS is licensed under CC BY-SA-NC

Figure 4. African Great rift valley system

At the start of your pull, the green part started to move away from each other, and the movement created a crack or rift. This will become deeper and will later on form a **rift valley**. One example of this is the so-called *Great Rift Valley system* that extends from the *Middle East to Mozambique*.

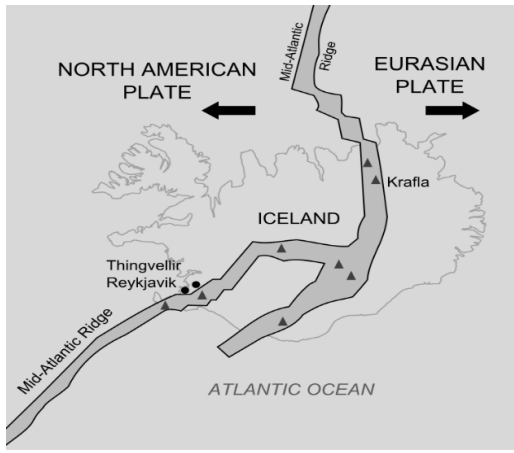
Since the plate is thick in the continents, these rifts will slowly widen alongside the formation of faults and rift valleys. When these rift valleys continue to move away from each other, water from different sources will start to come in and fill this rift valley. This is represented by the blue-colored part in the activity. It may start from a river or lake that contains freshwater drained to these deep areas of rift valleys. *Lake Baikal, Lake Superior,* and the *Dead Sea* are examples of these rift valley lakes. And once the crack extends and is filled with ocean water, this will start to form another body of water, for example, the *Red Sea* that lies between Africa and Arabia. And later, will be developed into an ocean.



Map of Baikal Lake by Sansculotte is licensed under CC BY-SA

Figure 5. Formation of Lake Baikal due to divergence plate boundary.

Molten materials under the oceanic crust undergo **convection currents** that assist the movement of the plates. As soon as the crack starts to form, the magma fills this up and since it is under the ocean, this will cool and harden which is characterized as **pillow lava**, thus, creating new crust. This process will continue creating a mountain system on the seafloor called a **mid-ocean ridge**. The most famous mid-ocean ridge is the **Mid-Atlantic Ridge**, which was developed as a result of the divergence of *Eurasian* and *North American*, and *African* and *South American Plate*s. In addition, when these ridges reach the surface of the ocean, a new volcanic island is formed and one example of this is Iceland**.**



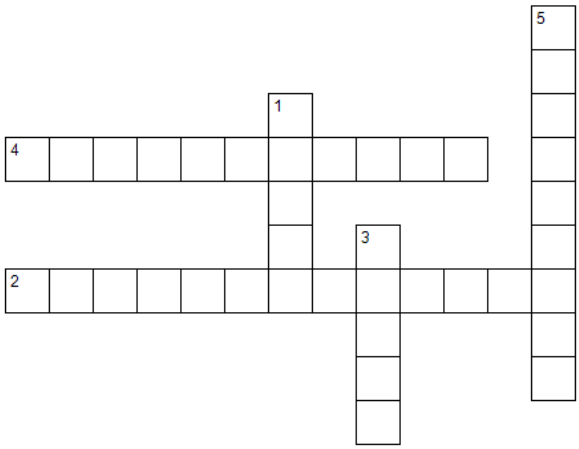
Major fractures zones of the Atlantic can be seen on this ocean depth map by *National Oceanic and Atmospheric Administration* is licensed under CC BY-SA

Iceland Mid-Atlantic Ridge map by *United States Geological Survey* is licensed under CC BY-SA

Figure 6. Mid Atlantic Ridge and Iceland as result of divergent plate boundary

What's More

Answer the crossword below.



Across:

2. mountain belt formed by pillow lava
4. depression in the land

Down:

1. molten material
3. outermost layer of the earth
5. moving away from each other

What I Have Learned

Complete the statement.

(1.) _____ plate boundary creates rift valleys and mid-ocean ridges, and in turn, creating new crust and new oceans. If the divergence happens in the continents, it forms (2.)_____, and if it happens on the seafloor it forms (3.)_____

As the continental plate moves (4.) _____ from each other, water from different sources will fill this gap. If it is under the ocean, (5.) _____ will fill the gap creating new seafloor.

What I Can Do

Draw a diagram of divergent plate boundaries and identify the different landforms formed due to this movement.

Assessment

Write TRUE if the statement is correct. If the statement is wrong, change the underlined word/s to make the sentence correct. Write your answers on a separate sheet of paper.

_____1. Divergent plate boundary movement forms a new ocean and new crust.
_____2. When continental plates move away from each other, ridges are formed.
_____3. The youngest crust is found in the middle of the ridges.
_____4. The Red Sea is an example of a body of water that was formed due to the divergence of two continental plates.
_____5. Eurasian Plate and South American Plate form the Mid-Atlantic Ridge.

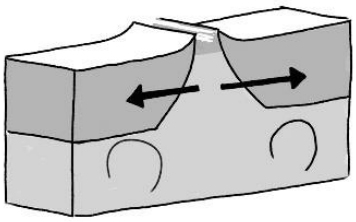
Additional Activities

Performance Task. Research on any famous geologic landform that was created by **divergent plate boundary movement** and create an infographic showing its, location, origin, and other interesting facts. It may be in a form of physical or digital output. Rubrics for scoring can be found on page 13.

Lesson 5	Processes along Transform Boundaries

What's In

Complete the diagram of what will happen in a divergent plate boundary after millions of years.



After millions of years?

(Describe what you have

What's In

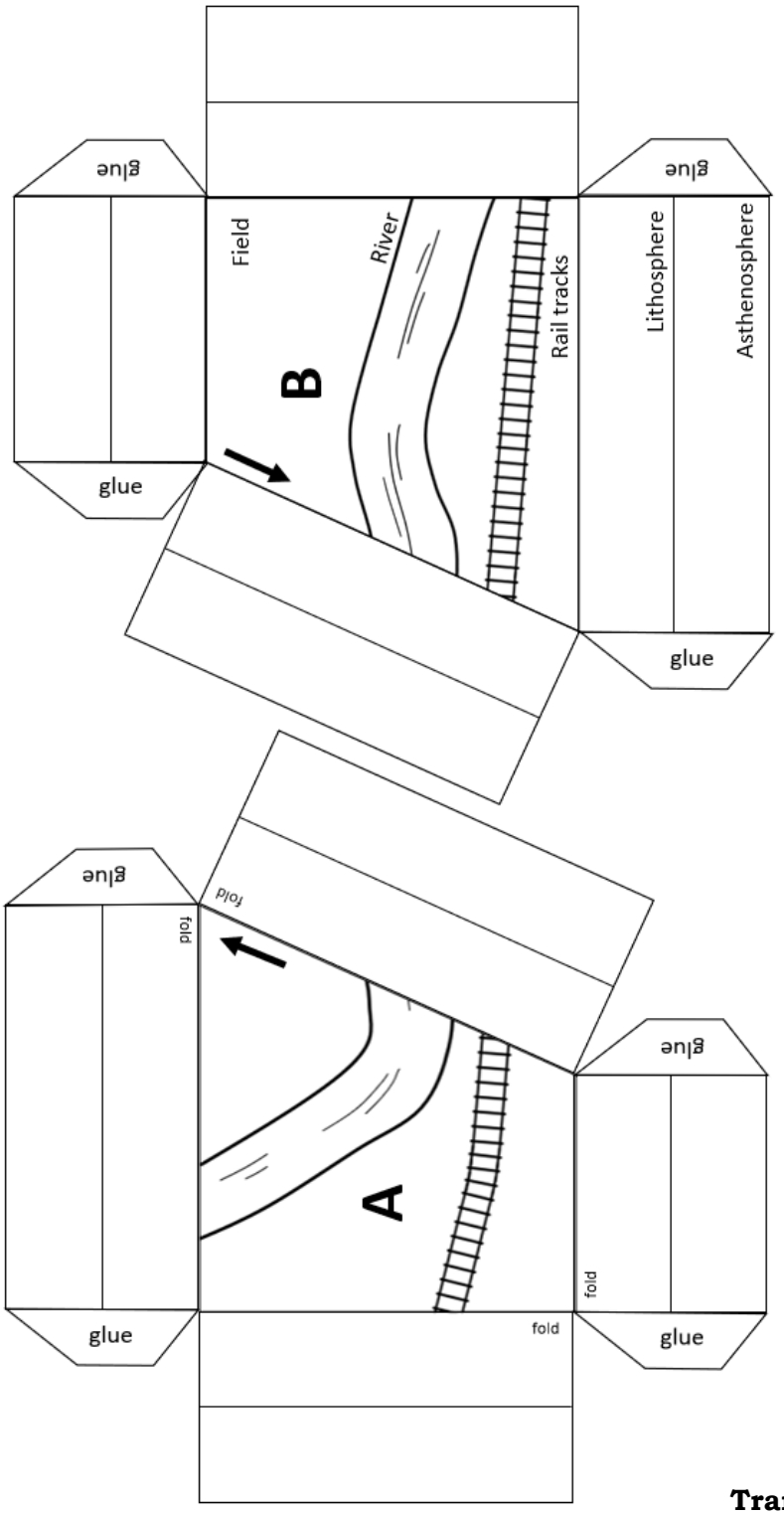
Activity: Sliding Plates

Objectives:

1. Explain the processes that occur along transform boundaries, and;
2. Identify the landforms associated with transform boundaries.

You Will Need:

copy of figure 7
pencil
scissors
glue
crayons
ruler



Legend:

Lithosphere – brown

Asthenosphere – Red

Field – Green

River - Blue

Figure 7

Transform Plate Boundary Model

Illustrated by Keith Angeline N. Alejandro

Procedure:

A. Constructing the transform plate boundary model

1. **Trace** Figure 7 on a short bond paper and color your work
   *(*Lithosphere – *Brown*, Asthenosphere – *Red,* Field – *Green*, River – *Blue)*

   Note: You may use other available colors you have in hand.

2. Cut the edges of the figure and glue each side

B. Observing the landform and process using the model

1. Hold plates A and B together.
2. Slowly slide both plates following the direction of the arrows.
3. Observe what happens. Write your observations and answers to the guide questions.

Guide Questions:

1. What interaction happens between the plates?

2. What will happen to the plates if the sliding continues to occur?

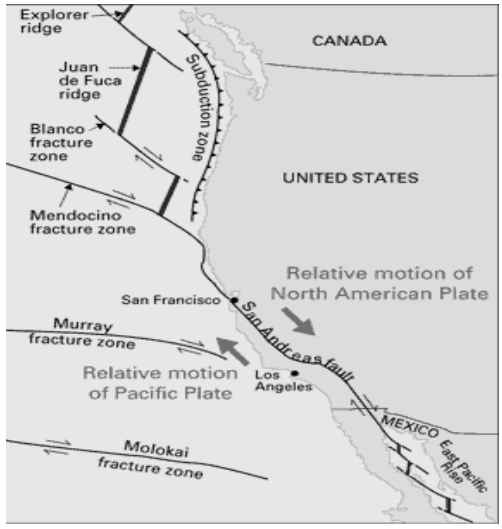
3. What events might occur during this process?

4. What landforms are formed?

What Is It

In Activity 5, you noticed that the plates may also slide from each other, they neither converge nor diverge,—this is called **transform boundary movement**. If this movement will continue to occur, deformation of the different land areas may occur like breakage of the river, movement of roads and rail.

Also, **faults** are formed by transform boundary movement. These are fractures in the lands that when the plates move it causes earthquakes. There are also transform faults under the oceans. Transform plate boundary movement assists in the processes and landforms that occur in convergent and divergent plate movement.



transform fault by USGS is licensed under CC BY-SA

Figure 5: San Andreas Fault formed by the relative motion of North American Plate and Pacific Plate

The most famous transform boundary is the *San Andreas Fault*. It is formed between the *North American plate* and *the Pacific plate*.

What's More

Place the following terms under **TRANSFORM** if it is related to transform boundary movement and if it is for the other plate boundary movement place it under **OTHERS**.

TRANSFORM	OTHERS		
		Fault	Earthquake
		Ridge	Slides past each other
		Mountain	Moves away

What I Have Learned

Complete the statement.

At a (1.) _____, the plates are moving past each other. Neither subduction nor creation of rift valleys and ocean ridges was formed. It only forms (2.) _____ that can build up pressure and when released, it causes (3.) _____.

What I Can Do

Watch the movie "San Andreas" movie about a massive earthquake in California and how the protagonist survived and rescued his daughter. Then, make a 2 – 3 paragraph reflection paper on how you can apply some of the survival tips presented in the movie. Also, cite some misconceptions you will find in the movie.

Rubrics for scoring

Content (10 pts) Accurate and detailed information is provided.

Organization (10 pts) Information are systematically organized, and it is easy for the reader to understand

Assessment

Write TRUE if the statement is correct. If the statement is wrong, change the underlined word/s to make the sentence correct.

_____1. Transform fault produces shallow and deep earthquakes.
_____2. Mountains are formed by a transform boundary movement.
_____3. San Andreas Fault is formed due to the movement of the Pacific Plate and North American Plate.
_____4. Deformation in the land area is caused by transform boundary movement.
_____5. Buildings and infrastructures are safe to build near a fault.

Additional Activities

Performance Task. Research on any famous geologic landform that was created by **transform plate boundary movement** and create an infographic showing its, location, origin, and other interesting facts. It may be in a form of physical or digital output.

Rubrics for scoring

Content (10 pts) Accurate and detailed information is provided.

Organization (5 pts) Information are systematically organized, and it is easy for the reader to understand

Visual Appeal (5 pts) Letter, colors, images, layouts, and other visual elements help in the expressing overall idea of the infographics

Citation (5 pts). All sources of information are cited using APA style.

Posttest

Read and understand each item carefully and encircle the letter of the correct answer.

1. What happens to an oceanic plate when it moves toward a continental plate?
   A. It moves over the continental plate.
   B. It moves under the continental plate.
   C. Both folds upward.
   D. Both folds downwards.

2. When does subduction occur in a convergent plate boundary movement?
   A. When the density of both plates is the same.
   B. When one plate is denser the other plate.
   C. When both plates are heated by the mantle.
   D. No subduction occurs.

3. What happens to the surface of the ocean when an oceanic plate moves towards another oceanic plate?
   A. It decreases its size.
   B. It becomes foggy.
   C. It increases its size.
   D. Nothing happens.

4. Which is formed when a continental plate converges with another continental plate?
   A. Fault
   B. Island Arc
   C. Mountain
   D. Volcano

5. Which is formed when an oceanic plate converges with another oceanic plate?
   A. Fault
   B. Ridges
   C. Rift valley
   D. Island Arc

6. Which is linked to the mid-ocean ridge?
   A. Pillow lava
   B. Ocean trench
   C. Deep earthquake
   D. Subduction zone

7. Which best describes the rocks in the mid-ocean ridge?
   A. It forms a shallow valley
   B. It is youngest in the middle of the ridge.
   C. The center is relatively cooler compared to the surrounding area.
   D. The rocks at the ridges change its composition when it moves away.

8. What geologic feature is produced by the divergence of the Eurasian plate and North American plate?
   A. African Rift Valley
   B. Lake Superior
   C. Lake Baikal
   D. Iceland

9. What body of water is associated with divergent plate boundary movement?
   A. Pacific Ocean
   B. Taal Lake
   C. Red Sea
   D. Marikina River

10. Which two plates formed the Mid-Atlantic Ridge?
   A. African Plate and Eurasian Plate
   B. African Plate and South American Plate
   C. South American Plate and Nazca Plate
   D. Eurasian plate and South American Plate

11. What do you call the crack on tectonic plates that are usually produced by transform boundary?
   A. Mountain
   B. Volcano
   C. Faults
   D. Island

12. Earthquake is characterized by shaking of the ground. Which boundary can produce earthquakes once they move?
   A. Convergent Boundary
   B. Divergent Boundary
   C. Transform Boundary
   D. Earthquake Boundary

13. Which is true about transform boundary movement?
   A. It can form mountains.
   B. It can cause a shallow and deep earthquake.
   C. After millions of years, this will form a new ocean.
   D. One example of a transform boundary is the Mid-Atlantic Ridge.

14. Which two plates formed San Andreas Fault?
   A. Pacific Plate and South American Plate
   B. North American plate and Nazca Plate
   C. North American Plate and Pacific Plate
   D. South American Plate and Nazca Plate

15. Which BEST explains plate movement?
   A. The gravitational pull of the moon
   B. The convection current in the mantle
   C. The difference in the density in the outer core
   D. The magnetic field produced in the outer core

Answer Key

What I Know	What's In	What's New
1. A 2. C 3. D 4. A 5. C 6. B 7. D 8. C 9. C 10. A 11. A 12. A 13. B 14. C 15. C	**Lesson 1** 1.Convergent boundary – moving towards each other 2. Divergent Boundary – moving away each other 3. Transform Boundary – Slide past each other Lesson 2 1. Subduction 2. Trench 3. Volcano Lesson 3 Continental oceanic – oceanic plate subducts under continental plate Oceanic oceanic – denser oceanic plate subducts under less dense oceanic plate Similarities – forms volcano, mountain, and trenches **Lesson 4** 1. Continental- oceanic (Nos. 2-4 can be interchanged) 2. Mountain 3. Volcano 4. Trench 5. Island arc 6. Continental-continental 7. Mountain ranges **Lesson 5** 1. Drawing should show body of water.	**Lesson 1** Depending on the orientation of the notebook. If they follow the picture in the module, The answer should be: 1. Right. Because the notebook's side that converges is more compact 2. Left. Because the notebook's side the converges is loose. 3. The Right notebook moves beneath the left notebook because it is more compact. 4. the plate will reach the mantle. 5. subduction, earthquake. 6. mountains, volcanoes, and trenches. **Lesson 2** 1. One notebook subduct under the other notebook. This is because the notebook is more compact than the other. 2. the plate will reach the mantle. 3. Subduction, earthquake, and tsunami. The landforms are undersea volcano, trench, mountains. **Lesson 3** 1. Both notebooks move upward because the sides that converges are loose. 2. the plates will continue to rise. 3. Earthquake. This will form mountain. **Lesson 4** 1. Green is the crust, blue is the ocean, red is the new crust (lava) 2. the notebook eventually changes its color from green to blue to red. 3. at the start, bodies of water. at the end, new crust. **Lesson 5** 1. The plates slides past each other. 2. The land will be deformed. River and rail road will be broken. 3. Earthquake 4. Faults or cracks in the land

What's More	What I Have Learned	Assessment
Lesson 1 1. Volcano 2. trench 3. OceanicPlate 4. Subduction 5. Continentalplate Lesson 2 1. Subduction 2. Trench 3. Volcanic island arc 4. Tsunami 5. Collision Lesson 3 Words that can be searched 1. Mountain 2. Collision 3. Earthquake 4. Folding 5. Everest 6. Himalayas Lesson 4 1. Magma 2. Midocean ridge 3. Crust 4. Rift valley 5. Divergent **Lesson 5** Transform Fault Earthquake slide past Each other Others Ridge Mountain Moves Away	Lesson 1 1. Landforms 2. Continental oceanic convergence 3. Subduction 4. Trench 5. Mountain/ volcano 6. Tsunami Lesson 2 1. Oceanic -oceanic convergence 2. Trench/ deep-sea trench 3. Volcano/ undersea volcano 4. Island arc Lesson 3 1. Continental-continental convergence 2. Mountain ranges 3. Light and less dense 4. Earthquakes Lesson 4 1. Divergent 2. Rift valley 3. Mis-ocean ridge 4. Away 5. magma Lesson 5 1. Transform Boundary 2. Faults 3. earthquakes	**Lesson 1** 1. True 2. deep 3. True 4. continental 5. Mt. Andes **Lesson 2** 1. older 2. True 3. True 4. True 5. Pacific plate under phil. Sea plate **Lesson 3** 1. True 2. shallow 3. mountain 4. True 5. True **Lesson 4** 1. True 2. rift 3. True 4. True 5. African and South America **Lesson 5** 1. True 2. Fault 3. True 4. True 5. Not safe

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